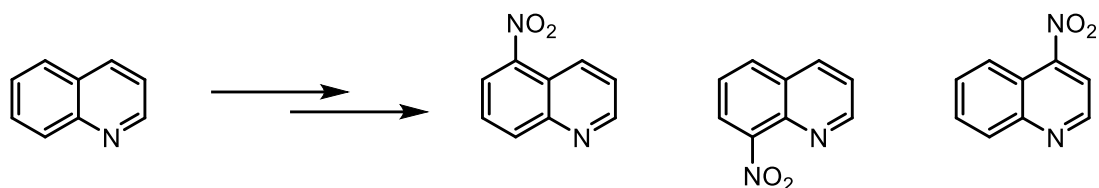
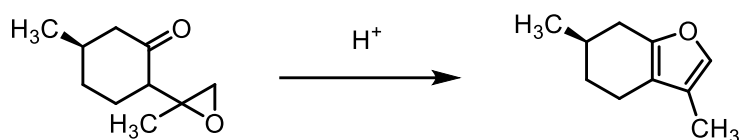


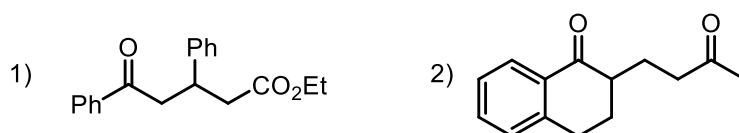
1. Suggest synthetic routes to achieve the following conversions.



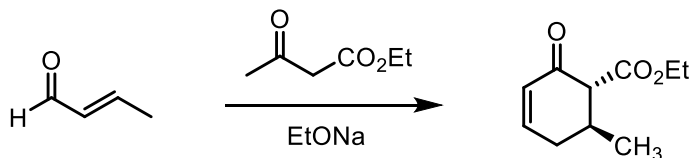
2. Suggest a mechanism for this reaction.



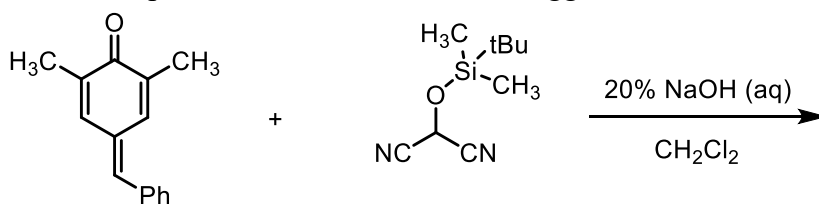
3. Suggest two different approaches to these compounds by conjugate addition of an enol(ate). Which do you prefer?



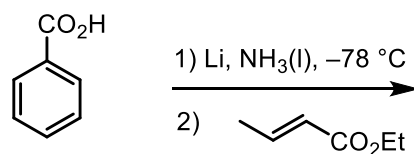
4. Suggest a mechanism for this reaction, commenting on any selectivity.



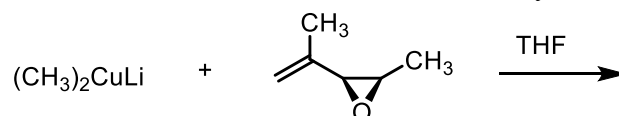
5. Predict the product for this reaction and suggest mechanisms.



6. Predict the product for this reaction and suggest mechanisms.



7. Predict the product for this reaction, and comment on any selectivity.



8. Propose a synthetic approach from toluene to this compound using conjugate additions.

